

INDIAN INSTITUTE OF TECHNOLOGY, KANPUR



MTH443: STATISTICAL AND AI TECHNIQUES IN DATA
MINING

A PROJECT REPORT ON
”BANKING CLUSTERING”

Submitted by :

Mrunal Dhiwar (221354)
Priya Yadav (221376)
Kumar Kanishk Singh (210544)
Kundan Kumar (210547)
Shabadpreet Singh (210954)

Under the Guidance of

Dr. Amit Mitra
(DEPARTMENT OF MATHEMATICS AND STATISTICS, IIT
KANPUR)

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1 INTRODUCTION

- *In the era of rapid technological advancements, data-driven decision-making has become essential across various sectors.*
- *The banking industry, in particular, has a wealth of data regarding customer transactions, credit usage, and spending behaviors, which, when analyzed effectively, can offer valuable insights into customer segmentation and targeted marketing.*
- *By leveraging statistical and artificial intelligence (AI) techniques, it is possible to explore complex patterns within this data, optimizing customer engagement and risk assessment strategies.*
- *This project focuses on applying clustering techniques to a bank's customer data to identify distinct customer groups based on their financial behaviors.*
- *The results of this analysis can assist banks in developing personalized services, understanding customer needs better, and implementing more precise risk management protocols.*

2 PROBLEM STATEMENT

The objective of this project is to perform clustering analysis on a banking dataset to segment customers based on their spending and repayment behaviors. With over 8,950 unique customer records encompassing 17 features such as credit usage, balance details, payment patterns, and purchase frequencies, the goal is to utilize clustering algorithms to categorize customers into distinct groups.

This segmentation can reveal behavioral patterns and help in identifying customer groups with shared characteristics, ultimately providing insights that can aid in enhancing customer relationship management, tailoring marketing strategies, and reducing credit risk.

3 DATASET DESCRIPTION

The dataset contains 8,950 entries and 17 features describing various aspects of customer spending and credit usage behaviors. Each row represents a unique customer, while the features provide insights into their credit card activity, balance, payment patterns, and more. The data will be used to perform clustering analysis to segment customers based on their spending and repayment behaviors.

Feature Overview

Below is a brief description of each feature:

- **CUST_ID**: Unique identifier for each customer.
- **BALANCE**: Current balance on the customer's credit card.
- **BALANCE_FREQUENCY**: Frequency of balance updates.
- **PURCHASES**: Total purchase amount by the customer.
- **INSTALLMENTS_PURCHASES**: Amount of purchases made in installments.
- **CASH_ADVANCE**: Cash advances taken by the customer.
- **PURCHASES_FREQUENCY**: Frequency of purchases.
- **ONEOFF_PURCHASES_FREQUENCY**: Frequency of one-off purchases.
- **PURCHASES_INSTALLMENTS_FREQUENCY**: Frequency of purchases in installments.
- **CASH_ADVANCE_FREQUENCY**: Frequency of cash advances.
- **CASH_ADVANCE_TRX**: Number of cash advance transactions.
- **PURCHASES_TRX**: Number of purchase transactions.
- **CREDIT_LIMIT**: Maximum credit limit for the customer.
- **PAYMENTS**: Total payments made by the customer.
- **MINIMUM_PAYMENTS**: Minimum payments made by the customer.
- **PRC_FULL_PAYMENT**: Percentage of months the full payment was made.
- **TENURE**: Tenure of the customer in months.

4 ANALYSIS

4.1 PRINCIPAL COMPONENT ANALYSIS

Introduction to PCA:

- Principal Component Analysis (PCA) is a dimensionality reduction technique widely used in data analysis and machine learning, particularly for high-dimensional datasets.
- It transforms the original features into a new set of uncorrelated components called principal components, ordered by the amount of variance they capture from the data.
- This method aids in identifying patterns, reducing noise, and making data more interpretable while preserving as much information as possible.
- In this project, the high-dimensional nature of the dataset required a reduction to facilitate more efficient clustering and analysis.
- After scaling the data using **StandardScaler** to standardize each feature to have a mean of zero and a standard deviation of one, PCA was applied.
- This approach ensured that features with larger scales did not dominate the variance captured by the principal components.

Results and Interpretation:

1)Principal Components:

- The PCA transformed the dataset into a series of **17 principal components** (PCA1 to PCA17).
- Each component represents a linear combination of the original features, capturing distinct variance portions.
- A subset of the transformed data is shown in the first table.

	PCA1	PCA2	PCA3	PCA4	PCA5	PCA6	PCA7	PCA8	PCA9	PCA10	PCA11	PCA12	PCA13	PCA14	PCA15	PCA16	PCA17
0	-1.684484	-1.070217	-0.469997	0.686725	-0.056053	-0.080315	-0.817358	-0.006871	0.119547	0.078525	-0.236376	0.051641	0.082369	0.188378	-0.149238	0.048147	0.000160
1	-1.132568	2.510222	-0.603830	-0.104391	-0.701246	1.074137	0.391847	0.196580	0.667960	0.778227	-0.870268	0.610110	0.033135	0.731434	0.533417	-0.082876	-0.000003
2	0.968926	-0.383446	-0.086175	1.252350	2.160177	0.394581	1.522617	-0.258764	-0.859574	0.001632	-0.762395	-0.683279	-0.697248	-0.042265	-0.221157	0.511270	-0.000023
3	-0.897248	-0.017764	-1.526942	1.031341	-0.189525	0.198574	0.210129	-0.761288	-0.014698	-0.393904	0.748709	-0.101937	0.112596	0.413100	0.084170	-0.014561	0.000013
4	-1.600519	-0.681375	-0.340010	1.024973	0.436712	-0.078247	-0.691393	0.263664	0.575308	0.122481	-0.455140	0.110116	0.036774	0.041101	-0.291180	-0.070558	0.000081

Figure 1: Figure 4.1.1

2) Explained Variance:

- The explained variance ratio for each component, shown in the second table, quantifies the **proportion of total variance each principal component captures**.
- The first few components capture most of the variance, indicating they hold the most critical information about the data.
- The **first component** (PCA1) captures approximately **27.3%** of the variance.
- The **second component** (PCA2) contributes an additional **20.3%**, bringing the cumulative explained variance to 47.6%.
- By the **fifth component**, over **70%** of the total variance is captured, suggesting that these first few components can represent the dataset effectively with minimal information loss.

	Eigenvalues	Explained Variance	Cumulative Explained Variance
Principal Component			
1	4.641399	2.729930e-01	0.272993
2	3.454852	2.032039e-01	0.476197
3	1.500345	8.824573e-02	0.564443
4	1.271731	7.479937e-02	0.639242
5	1.054981	6.205077e-02	0.701293
6	0.974672	5.732724e-02	0.758620
7	0.830031	4.881988e-02	0.807440
8	0.735091	4.323579e-02	0.850676
9	0.642875	3.781196e-02	0.888488
10	0.523674	3.080091e-02	0.919288
11	0.403382	2.372569e-02	0.943014
12	0.301515	1.773422e-02	0.960748
13	0.242932	1.428852e-02	0.975037
14	0.206852	1.216639e-02	0.987203
15	0.172152	1.012545e-02	0.997329
16	0.045405	2.670581e-03	0.999999
17	0.000012	6.852429e-07	1.000000

Figure 2: Figure 4.1.2

3) Cumulative Variance Plot:

- The cumulative variance plot further illustrates the **rate at which variance is captured**.
- With **approximately 90% of the variance explained by the first nine components**, this subset can be selected for further clustering and analysis, balancing data reduction and information preservation.

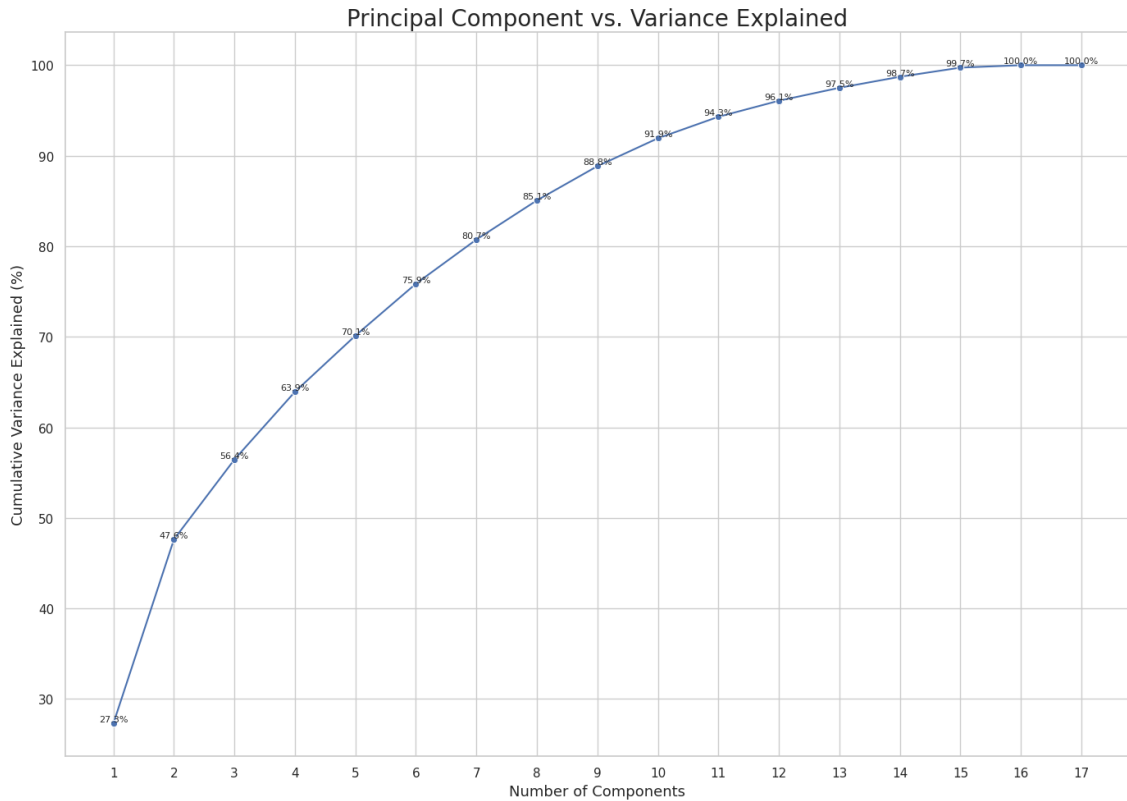


Figure 3: Figure 4.1.3

Conclusion:

- I. The application of Principal Component Analysis (PCA) in our project has significantly optimized both the efficiency and effectiveness of our data analysis.
- II. By transforming the original 17-dimensional dataset into a set of principal components, we have effectively reduced dimensionality while preserving the most critical information.

Key findings from the PCA include:

1. Dimensionality Reduction:

We reduced the dataset from **17 original features to 9 principal components**, a reduction of approximately 47% in dimensionality.

2. Variance Captured:

- PCA1 captures 27.3% of the total variance.
- PCA2 adds 20.3%, bringing the cumulative variance to 47.6% with just two components.
- PCA5 increases the cumulative variance captured to over 70%.
- By *PCA9*, we *capture approximately 90%* of the total variance.

3. Efficiency Gains:

The reduction in dimensions enhances computational efficiency, reducing processing time and resource consumption during clustering and further analysis.

4. Information Preservation:

- Despite the dimensionality reduction, the selected principal components retain the vast majority of the dataset's variability, ensuring that critical patterns and relationships are maintained.
- By focusing on these nine principal components, we strike an optimal balance between simplifying the dataset and preserving essential information.
- This streamlined dataset facilitates more efficient and potentially more accurate clustering, which is crucial for effective bank customer segmentation.

4.2 K-MEANS CLUSTERING

Introduction to K-Means Clustering:

- K-Means is an unsupervised machine learning algorithm that aims to partition data into a predefined number of clusters (K).
- Each data point is assigned to the cluster with the nearest mean, serving as a prototype of the cluster.

Results and Interpretation

- Selecting the appropriate number of clusters (K) is critical, as it impacts the quality of clustering.
- In our analysis, the optimal K was determined using methods like:
 - **Elbow Method:** We plot the Sum of Squared Errors (SSE) for each K value, and pick the point where the rate of decrease sharply slows, indicating the "elbow."

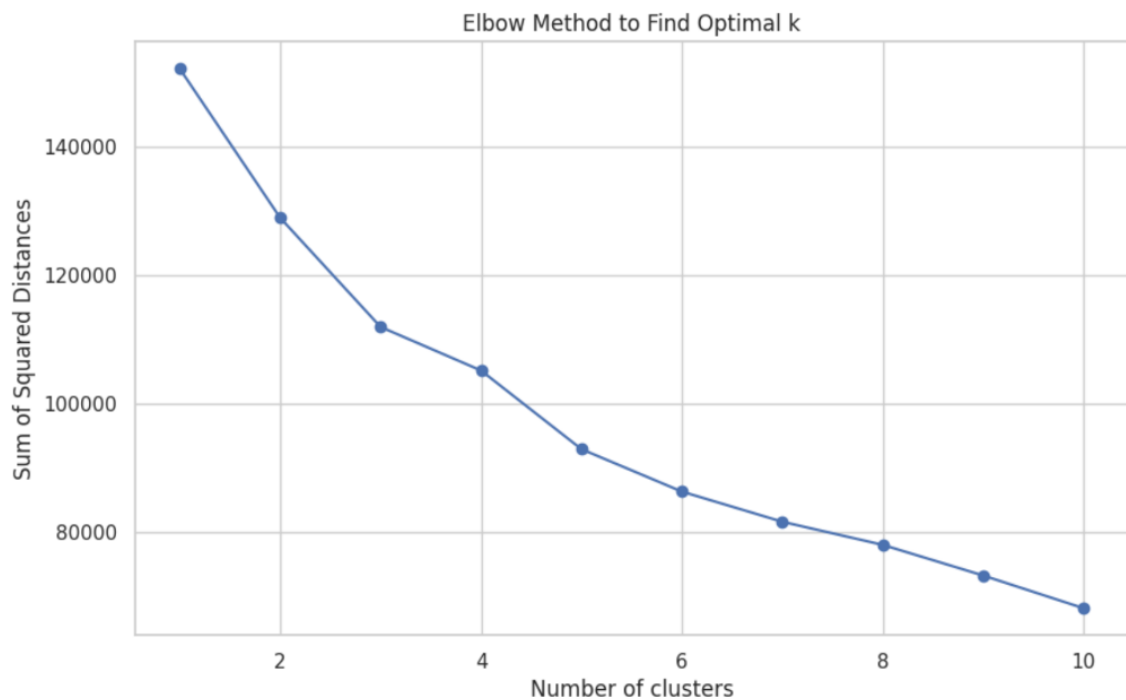


Figure 4: Figure 4.2.1

We can observe the sharp change in curve around $k = 4$ or 5 in figure 4.2.1. So, we can take either $k = 4$ or 5 .

- **Silhouette Score:** Measures how similar each point is to its cluster compared to others, with values closer to 1 indicating well-defined clusters. The silhouette score for $k=4$ and 5 are **0.22415** and **0.2099** respectively. The silhouette score is slightly higher for 4 clusters than for 5 clusters,

indicating that the **4-cluster** solution provides slightly better-defined clusters in terms of separation and cohesion.

- We now visualize clusters created by the K-Means algorithm on data that has been reduced to two principal components (PCA1 and PCA2). Each point in the scatter plot represents an observation from the dataset, and each cluster is shown in a distinct color. This visualization helps us by plotting clusters along two principal components, we can visually assess how well-separated the clusters are in reduced dimensional space.

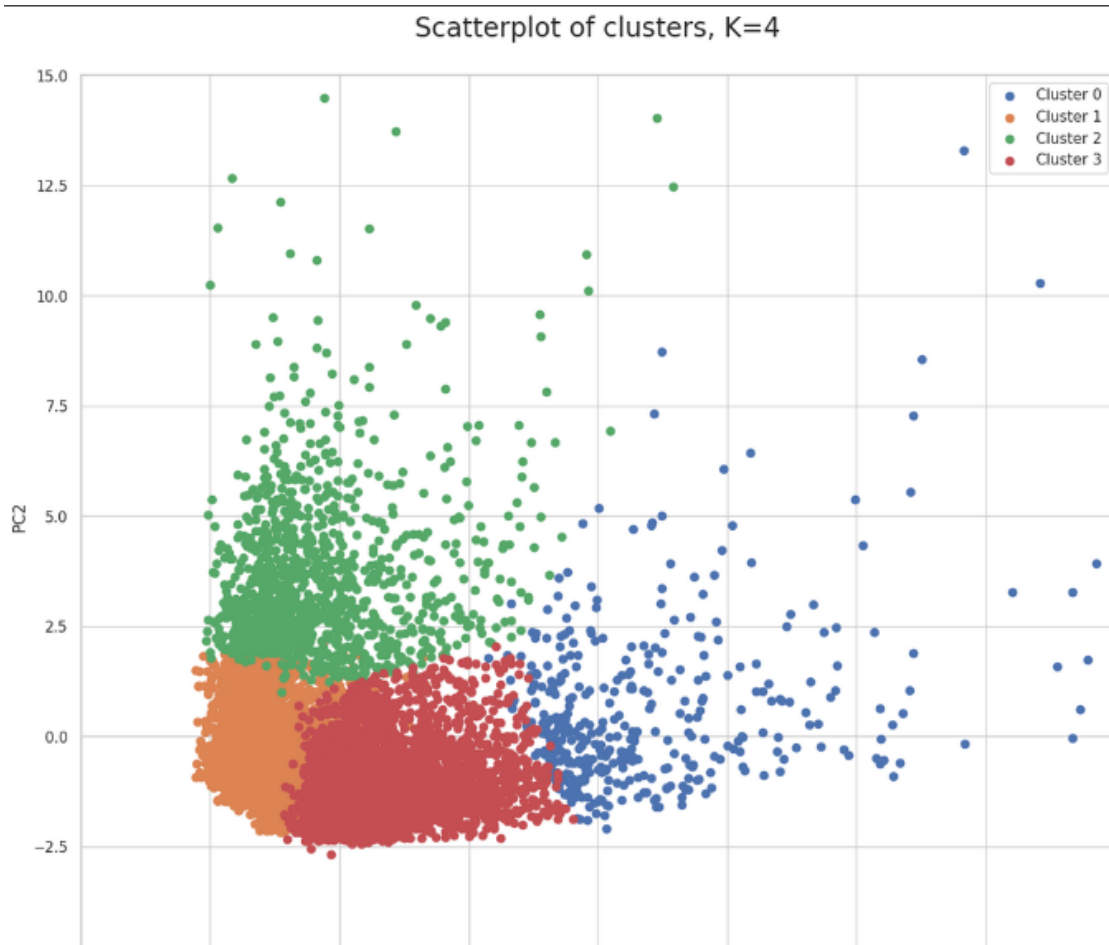


Figure 5: Figure 4.2.2

We can infer the following from the figure 4.2.2:

- **Cluster Separation:** The four clusters are generally well-separated, with Cluster 0 (blue) distinctly isolated on the right, while Clusters 1 (red) and 2 (green) overlap slightly, suggesting some similarity.
- **Density and Spread:** Cluster 3 (orange) is compact, indicating tightly grouped data points, while Clusters 1 and 2 cover more area, showing greater spread.

- **Principal Components:** The plot confirms that PC1 and PC2 capture significant data variation, effectively differentiating the clusters.

Lets take a look at the biplot in figure 4.2.3:

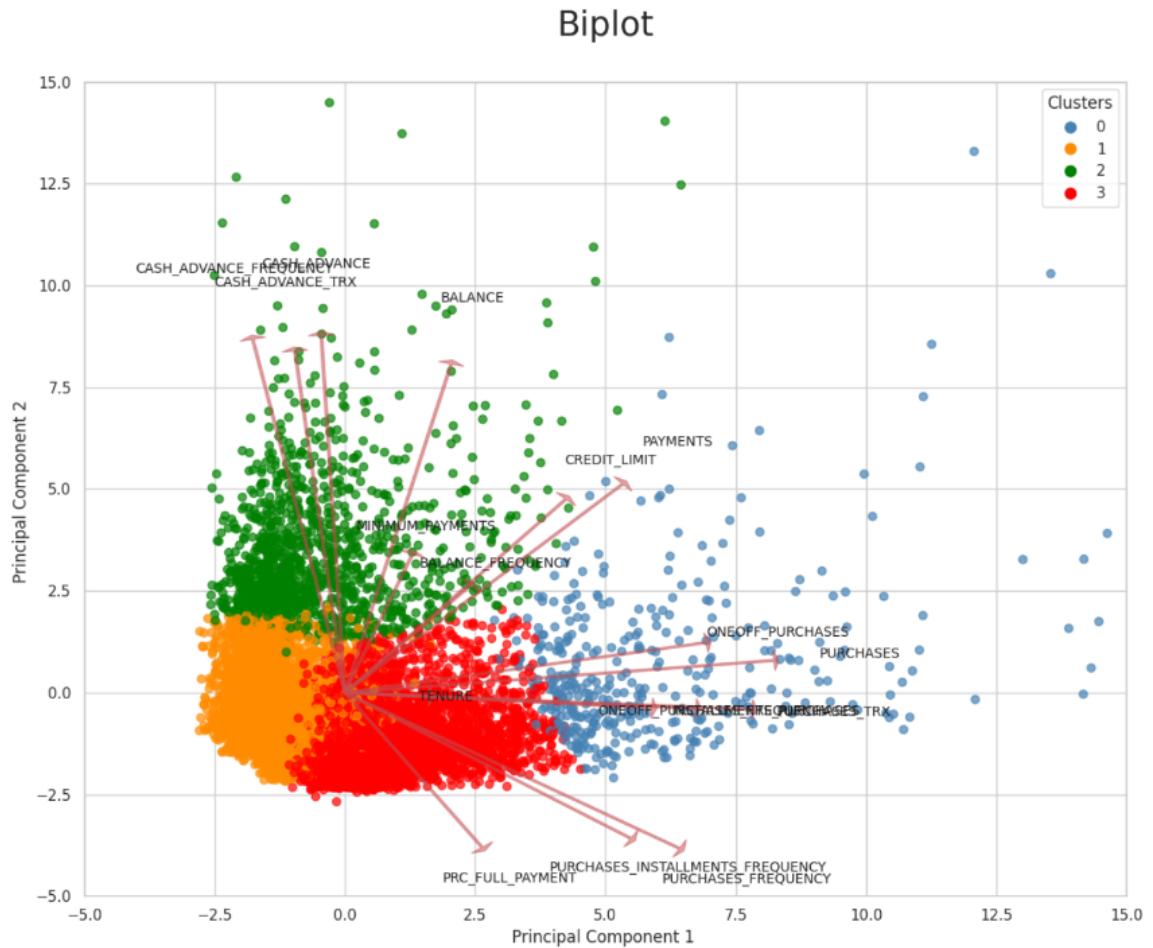


Figure 6: Figure 4.2.3

- **Feature Vectors (Arrows):** CASH_ADVANCE, CASH_ADVANCE_TRX, FREQUENCY: Strongly aligned with PC2 and Cluster 2 (green), indicating high cash advance usage and frequency.
- **TENURE, PRC_FULL_PAYMENT:** Point towards Cluster 3 (red), suggesting this group includes customers with long tenure and consistent full payments.
- **ONEOFF_PURCHASES, PURCHASES, PURCHASES_FREQUENCY:** Aligned with PC1 and Cluster 0 (blue), indicating frequent purchase behaviors.
- **BALANCE, CREDIT_LIMIT, MINIMUM_PAYMENTS:** Associated with Cluster 2 (green), indicating higher balances and credit limits.
- **Cluster Characteristics:**

- * Blue Cluster (Cluster 0): Represents frequent purchasers with high transaction activity.
 - * Green Cluster (Cluster 2): Defined by high balances, credit limits, and cash advances.
 - * Red Cluster (Cluster 3): Represents loyal, long-tenure customers with consistent payments.
 - * Orange Cluster (Cluster 1): Likely lower credit activity or engagement.
- Instead of fitting K-means on the entire dataset, we now apply it to the first 10 principal components, as they capture a significant amount of variance, which helps reduce computational cost. We then use the elbow method to identify the optimal number of clusters.

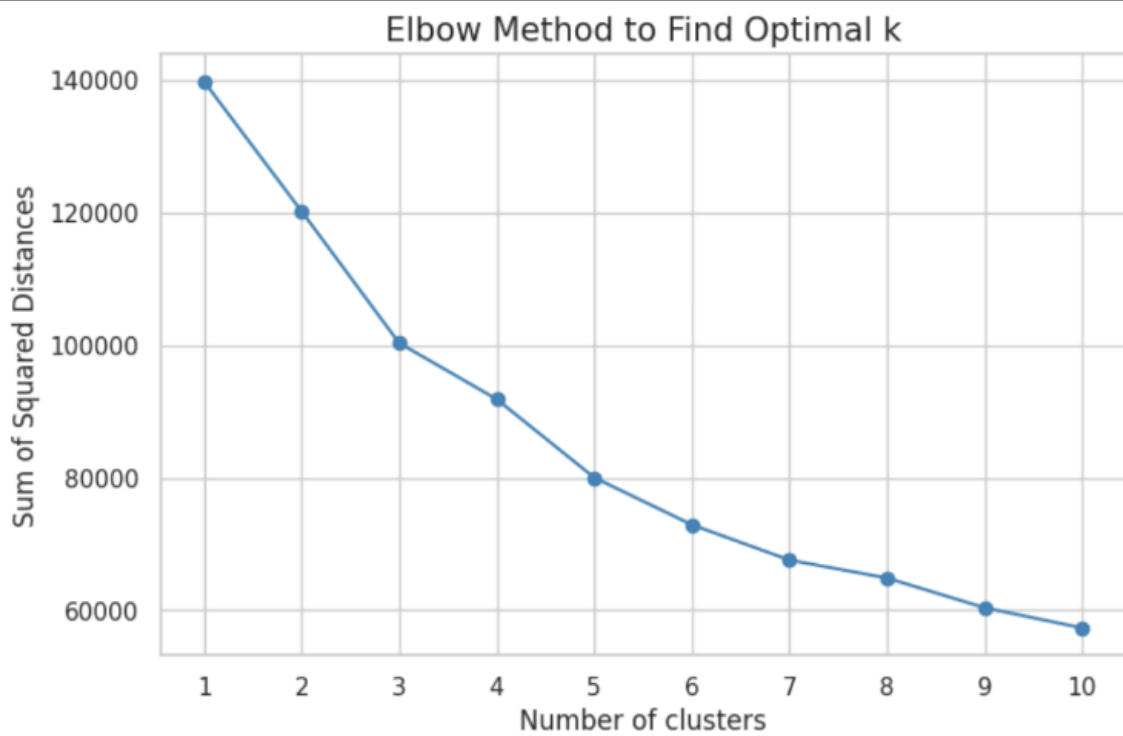


Figure 7: Figure 4.2.4

We can see from figure 4.2.4 that the optimal number of cluster is 4.

We can see clearly from figures 4.2.5 and 4.2.6 that both the methods give us the same result.

4.3 HIERARCHICAL CLUSTERING

Hierarchical Clustering

Hierarchical clustering is a type of unsupervised machine learning algorithm used for grouping data points into clusters based on their similarity. Unlike other clustering methods, hierarchical clustering creates a tree-like structure called a *dendrogram*, where each level represents clusters merged at different similarity levels. This technique is particularly useful for visualizing the nested structure of data and identifying natural groupings.

Types of Hierarchical Clustering

There are two primary types of hierarchical clustering:

- **Agglomerative Hierarchical Clustering:** This is a "bottom-up" approach where each data point starts as its own cluster. At each step, the two closest clusters are merged until a single cluster encompassing all data points is formed. Agglomerative clustering is the most commonly used form of hierarchical clustering.
- **Divisive Hierarchical Clustering:** This is a "top-down" approach where all data points start in a single cluster. At each step, the algorithm splits the most dissimilar clusters until each data point is in its own cluster.

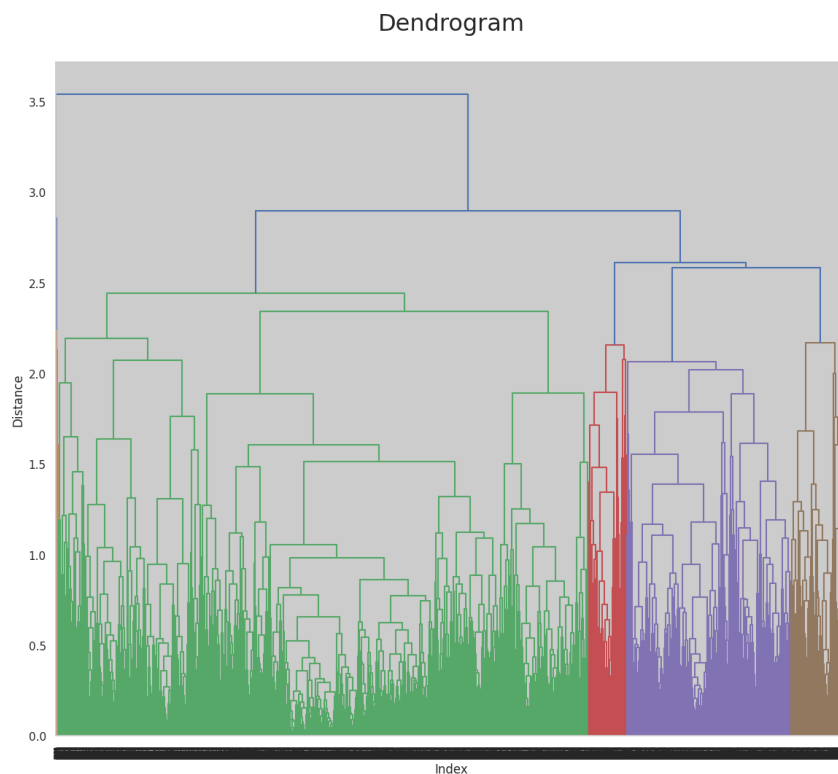


Figure 8: Fig. 4.3.1

In this project, hierarchical clustering was applied to banking data to identify clusters of similar entities. The dendrogram above represents the hierarchical relationships among data points, with branches indicating the similarity or dissimilarity between points based on a chosen distance metric. The height of each link shows the distance at which clusters merge, and the color coding highlights different clusters formed at a specific distance threshold. This approach helps reveal natural groupings in the data, which can provide insights into customer segmentation, product targeting, or risk assessment in the banking sector.

4.4 GAUSSIAN MIXTURE MODEL

Introduction to Gaussian Mixture Models

- Gaussian Mixture Models (GMMs) are a **probabilistic clustering approach** that assumes data points are generated from a mixture of several Gaussian distributions, each representing a cluster.
- Unlike KMeans, which forms spherical clusters with hard assignments, **GMM assigns data points to clusters based on probabilities**, allowing more flexible, elliptical cluster shapes.

Results and Interpretation

- We optimized the hyperparameter of Gaussian Mixture Model (GMM) using Optuna for maximizing silhouette score.
- the optimized silhouette score of 0.336 indicates good clustering quality.
- We can see from figure 4.3.1, clusters are better separated, though some overlap remains, particularly in the center.

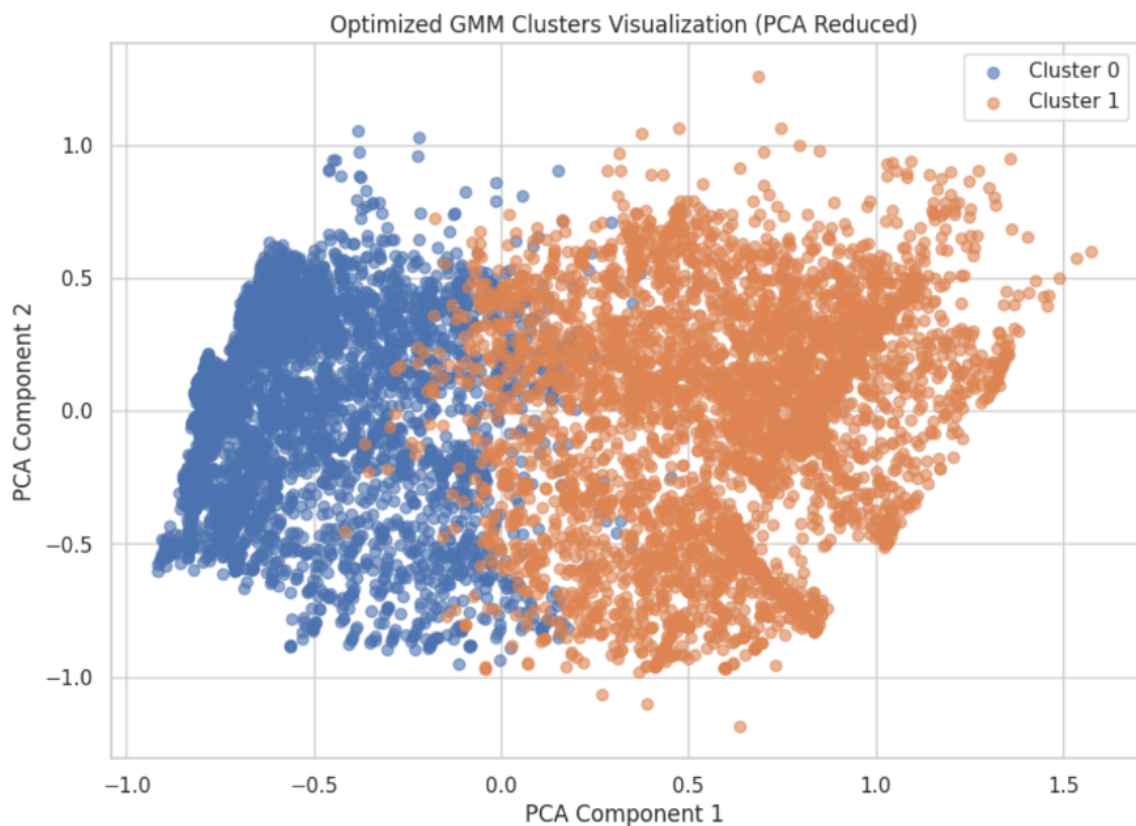


Figure 9: Figure 4.4.1

- The clusters have an elliptical shape, which is good, but the overlap indicates that the clusters aren't as well-defined and separate as GMM ideally assumes. This could impact the effectiveness of the model, especially if the goal is to have distinct, non-overlapping clusters.
- Let's check assumptions of GMM it might possible that assumptions not satisfied leads to poor performance. We check for independence of features within each cluster:

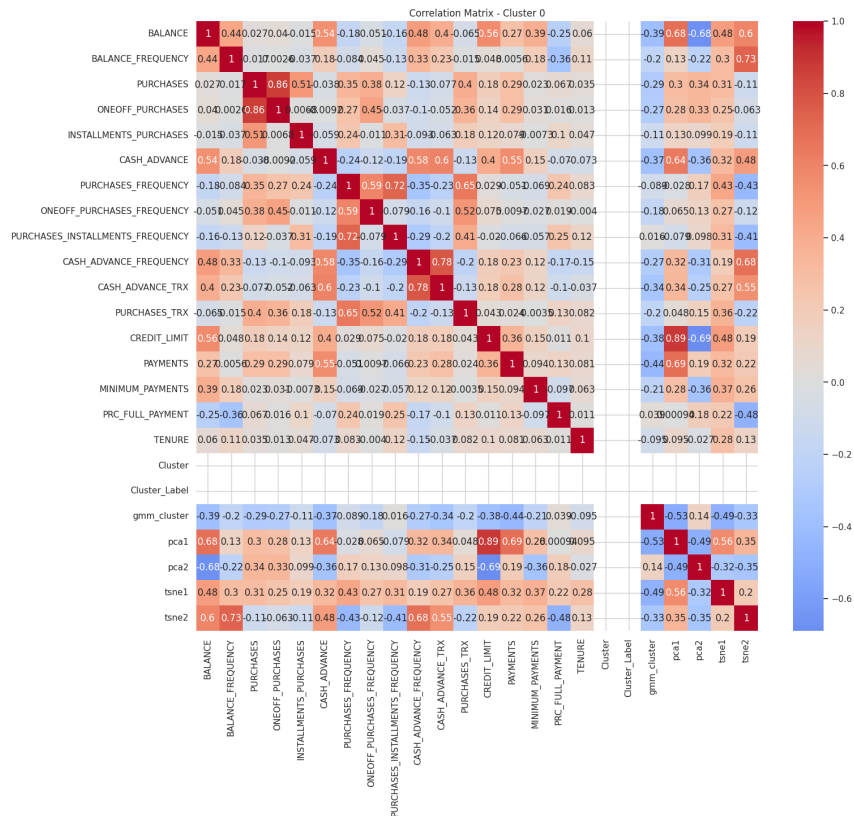


Figure 10: Figure 4.4.2

The correlation matrices for both clusters (figures 4.4.2 & 4.4.3) show that many features are not independent within each cluster. This is especially notable for features directly related to similar aspects (e.g., cash advances, payments, and purchases), which tend to be correlated due to their inherent relationships in the data.

Given this, the independence assumption in GMM is not fully met, especially if you are using spherical or diag covariance types, which assume uncorrelated features within each cluster.

- Cluster Weights:

Cluster Proportions based on Data:

Cluster0: 0.528774

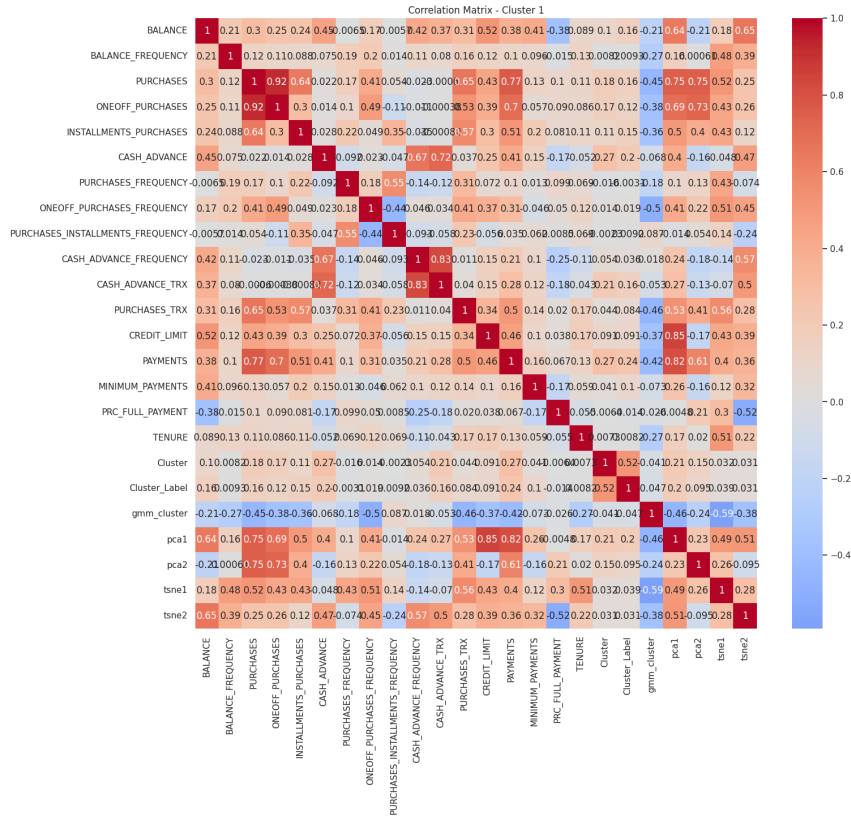


Figure 11: Figure 4.4.3

Cluster1: 0.471226

GMM Estimated Weights:

Cluster0: 0.52957806

Cluster1: 0.47042194

The estimated weights and actual proportions align closely. This suggests that GMM is effectively modeling the underlying distribution in terms of cluster sizes.

- The poor performance of GMM, reflected in the low silhouette score, is likely due to the fact that not all of its assumptions are fully satisfied in this dataset. When key assumptions—such as normal distribution of each cluster, feature independence, and clear separation of clusters—are not met, the model may struggle to accurately capture the data structure, leading to suboptimal clustering results.

5 CONCLUSION

- After applying and evaluating K-Means, Gaussian Mixture Model (GMM), and Hierarchical Clustering on the dataset, K-Means emerged as the most effective clustering method, providing the highest silhouette score and well-separated clusters. K-Means achieved optimal results with 4 clusters, which aligned well with the natural grouping in the data, as confirmed by both visualizations and cluster compactness.
- Unlike GMM and Hierarchical Clustering, which showed considerable overlap and imbalanced cluster sizes, K-Means created distinct and balanced clusters, demonstrating its robustness and suitability for this dataset.
- Its efficiency, simplicity, and ability to form compact clusters with clear boundaries make K-Means the best choice for this analysis.

6 REFERENCES

MTH443A : Statistical and AI Techniques in Data Mining